

Lecture 5

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1 Encoding Linear Codes

Consider q^k codewords and q^k messages. Denote each message μ as an n -tuple $(\mu_1 \mu_2 \dots \mu_n)$, where each $\mu_i \in \text{GF}(q)$. One way to encode the message μ is

$$c = \mu \mathbf{G},$$

where $\mathbf{G}$ is the generator matrix and c is the corresponding codeword. Let us denote a code by (n, k) where n is the length of the codeword and k is the dimension of the linear code $\mathbf{C}$.

Example 1: For a simple parity check $(3, 2)$ code, let

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Therefore the message $\mu = [0 \ 1]$, is encoded as

$$c = [0 \ 1] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = [0 \ 1 \ 1]$$

similarly for $\mu = [1 \ 1]$, we get $c = [1 \ 1 \ 0]$

Example 2: Let us now consider a $(7, 4)$ Hamming code whose generator matrix is

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Just as a reminder, we get the dimensions of $\mathbf{G}$ as

$$(\text{Dimension of the code } \mathbf{C}) \times (\text{Length of each codeword})$$

On close inspection, this matrix has a **systematic** form $[I|A]$ where I is a 4×4 identity matrix and A is a 4×3 matrix. Using this matrix, our codeword would be of the form [message bits | parity bits]. Let's say our message is $\mu = [\mu_0 \ \mu_1 \ \mu_2 \ \mu_3]$, and our codeword is $c = [c_0 \ c_1 \ c_2 \ c_3 \ c_4 \ c_5 \ c_6]$. Then we will get

$$c_i = \mu_i, i \in \{0, 1, 2, 3\} \quad (1)$$

$$c_4 = \mu_0 + \mu_2 + \mu_3 \quad (2)$$

$$c_5 = \mu_0 + \mu_1 + \mu_2 \quad (3)$$

$$c_6 = \mu_1 + \mu_2 + \mu_3 \quad (4)$$

We prefer to use systematic generator matrix as the original message bits can be easily recovered from the codeword (this is done when the error rate of the channel is very low, for example), instead of solving the linear combinations if we had some other form for $\mathbf{G}$. It is always possible to convert a non systematic matrix to a systematic matrix

2 Parity Check Matrix

Let $\mathbf{C}$ be a subspace, with $\dim(\mathbf{C}) = k$. We know that there exists $\mathbf{C}^\perp$, an orthogonal subspace to $\mathbf{C}$, which will also have a corresponding generator matrix. It is this matrix that is called the Parity Check matrix, denoted by $\mathbf{H}$.

$\mathbf{H}$ is a $(n - k) \times n$ matrix, which satisfies the following property

$$\mathbf{H}\mathbf{c}^T = \bar{0},$$

which further implies that

$$\mathbf{H}\mathbf{G}^T = \mathbf{O}$$

If we have a generator matrix which is a systematic matrix, its parity check matrix will be of the form $[-A^T | I]$.

Example 3: For the (3,2) simple parity check code, parity check matrix can be given by

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

It is interesting to note that the dual of an $(n, n - 1)$ simple parity check code comes out to be a $(n, 1)$ repetition code!

Example 4: For the (7,4) Hamming code, the parity check matrix will be

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

One can verify that for any codeword $\mathbf{c}$ in $\mathbf{C}$, $\mathbf{H}\mathbf{c}^T = \bar{0}$ is satisfied.

2.1 Minimum weight in terms of $\mathbf{H}$

Just like we characterized the minimum weight of a code in terms of its Hamming distance, we can do it in terms of the parity check matrix as well.

Claim: If $\mathbf{H}$ is the parity check matrix of code $\mathbf{C}(n, k, d)$, then, minimum distance d is the largest integer such that any $(d - 1)$ columns of $\mathbf{H}$ are linearly independent.

Proof: We prove this claim in two parts:

1. There exist at least d linearly dependent columns in $\mathbf{H}$.

2. Any collection of $(d - 1)$ columns in $\mathbf{H}$ is linearly independent.

Part 1: Let $\mathbf{H}$ be the $(n - k) \times n$ parity check matrix. Let each codeword in $\mathbf{C}$ be of the form $c = (c_0, c_1, \dots, c_{n-1})$. Then, if we think of $\mathbf{H}$ as being composed of n columns like this:

$$\mathbf{H} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ h_0 & h_1 & \dots & h_{n-1} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$\mathbf{H}c^T = \bar{0}$ implies that

$$\sum_{j=0}^{n-1} c_j h_j = 0.$$

Thus, the linear combination of the columns of $\mathbf{H}$, with any codeword c , evaluates to 0.

Therefore, if we are given a codeword, we can identify which columns of $\mathbf{H}$ are linearly dependent. How? The columns of $\mathbf{H}$ corresponding to non-zero entries in c are linearly dependent. We know that if the minimum distance of a code is d , then its Hamming weight (also the minimum number of non-zero entries in any codeword) is d . Now, since there must exist at least one codeword with Hamming weight d , there will be at least one set of d columns in $\mathbf{H}$ which are linearly dependent.

Part 2: Let's assume to the contrary that there exists a collection of $(d - 1)$ columns of $\mathbf{H}$ that are linearly dependent. Then there must exist a codeword in $\mathbf{C}$ such that $w(c) = d - 1$, i.e., a vector with only $(d - 1)$ non-zero entries since,

$$\sum_{j=1}^{n-1} c_j h_j = 0$$

The above coefficients form a new codeword which has $(d - 1)$ non-zero entries. But this implies that the minimum weight is $(d - 1)$, which further implies that the minimum distance is $(d - 1)$. However, this is false, as our code $\mathbf{C}$ has minimum distance d . Thus, we arrive at a contradiction.

Hence proved.

3 Decoding of linear codes

Consider a (n, k, d) linear code. Let c be the transmitted codeword and r be the received codeword.

$$r = c + e$$

where e is the error introduced in the channel. We partition $\mathcal{F}^n$ vector space into q^k decoding subspaces $D_1, D_2, \dots, D_{q^k}$. If r lies in D_i then the decoded codeword is c_i . Linear

code helps us in simplifying the process. We use **Coset Decomposition**.

$$\begin{bmatrix} c_0 & c_1 & c_2 & \dots & c_{q^k} \\ e_1 & e_1 + c_1 & e_1 + c_2 & \dots & e_1 + c_{q^k} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ e_{q^{n-k}} & e_{q^{n-k}} + c_1 & e_{q^{n-k}} + c_2 & \dots & e_{q^{n-k}} + c_{q^k} \end{bmatrix}$$

Here we choose $c_0 = 000\dots 0$.

In this case, each column is a decoding subspace and each received vector r lies in one of these subspaces. The codeword corresponding to that subspace is our decoded codeword.

Claim: We know $r = c_i + e$, if the error pattern e is a coset leader, then we correctly decode the codeword and if e is not a coset leader then we will incorrectly decode the codeword

Proof: e and r both lie in the same coset as $r = c_i + e$. If e is a coset leader then c_i would be the decoded codeword as it is how we have defined.

If e is not a coset leader then we can define

$$e = e_l + c_j$$

where e_l is the coset leader of the coset in which r lies and c_i is a codeword.

$$r = c_i + e \tag{5}$$

$$\implies r = c_i + e_l + c_j \tag{6}$$

$$\implies r = e_l + (c_i + c_j) \tag{7}$$

$$\implies r = e_l + c_k \tag{8}$$

where $c_k \neq c_i$. Therefore we incorrectly decode the codeword to c_k . Hence proved

Now the important question remains, how are we supposed to choose the coset leaders?

We use the **Standard Array Procedure**. It states that at each row i , we choose the lowest available weight vector as the coset leader.

Claim: Using Standard Array Procedure, we get the nearest neighbor decoding

Proof: Consider

$$r = c_i + e_l$$

where e_l is the l^{th} coset leader. This will decode r to c_i . Since we have linear codes

$$d(r, c_i) = w(e_l) \tag{9}$$

$$d(r, c_j) = d(c_i + e_l, c_j) = w(e_l + c_i - c_j) \tag{10}$$

$$\text{Let } c_i - c_j = c_k \tag{11}$$

$$\text{By definition: } w(e_l + c_k) \geq w(e_l) \tag{12}$$

Hence proved.

Claim: For any (n, k, d) linear code and $t = \lfloor \frac{d-1}{2} \rfloor$, all n -tuples of weight $\leq t$ are coset leaders in a standard array.